12-06 第1篇

阅读理解（说明文） Why Do We Feel Sleepy in the Afternoon?

① At roughly **14:20** on most ordinary school days, something quietly strange happens in classrooms across the world. A Grade 8 student, who was wide awake at 11:00, suddenly feels her eyelids become very heavy. The teacher is drawing a kind, clear diagram on the board. The student is not bored. She did not stay up late. She had a reasonable lunch at 12:00. Yet, between 14:00 and 15:30, her brain seems to be quietly pulling her downward, toward the desk. This short, honest daily sleepiness has a name: **the post-lunch dip**. For a long time people blamed lunch itself. It turns out lunch is only a small part of the story. The real reason sits deep inside our biology: **we are built to slow down in the early afternoon, whether we have eaten or not.**

② The central cause is the **circadian rhythm** — a roughly 24-hour internal clock that every cell of the human body follows. This clock, sitting in a tiny brain region called the **suprachiasmatic nucleus (SCN)** just above where the two optic nerves cross, controls alertness, body temperature, hormones, and sleep. Hundreds of studies, going back to the 1980s, have measured the clock's natural alertness curve. It is **not a straight line**. It has **two dips per day**: the famous deep one at around **3:00 — 4:00 in the morning**, and a second, smaller one at around **14:00 — 16:00 in the afternoon**. Both dips exist in people who work night shifts, in people who skip lunch completely, and even in people isolated for months in Arctic caves. The afternoon dip, in other words, is a fixed feature of our species, not a mistake caused by rice.

③ **But lunch plays a supporting role.** After a medium-sized lunch, blood flow shifts partly toward the stomach and intestines (needed for digestion). Insulin rises to help cells absorb sugar. A natural small rise in **tryptophan**, an amino acid used by the brain to make the sleep-related chemical **serotonin**, follows a carbohydrate-rich meal — this is why a big bowl of white rice or noodles makes us, honestly, a little sleepier than a light salad. A 2022 meta-study of 18 experiments (University of Loughborough, UK) found that a lunch heavy in refined carbohydrates deepened the afternoon dip by **about 20%**, while a light lunch reduced but **did not remove** the dip. The dip was there either way. Lunch turns the volume up, not on.

④ **Body temperature is another part of the answer.** Human core temperature rises and falls across the day — about 0.5 °C between its highest and lowest. Alertness strongly tracks temperature: when the body warms up, we feel sharp; when it cools slightly, we feel heavy. In most people, there is a **small dip in core temperature around 14:00 — 15:00**, even in very cool rooms, even after a workout. The brain reads this small cooling as "low-energy time" and reduces alertness accordingly. On very cold winter afternoons,

the dip can feel stronger — one reason December 2:30 feels longer than June 2:30 in many schools.

⑤ **Teenagers have a twist.** The circadian rhythm of humans aged 13—20 runs **about 1 — 2 hours later** than it does in adults. This means that a teenager's peak alertness is pushed to mid-morning and her afternoon dip can be **deeper and earlier** than in a forty-year-old. A 2023 study of 1,740 Chinese middle-school students (Beijing Normal University) confirmed the finding: **73%** of Grade 7—9 students reported significant sleepiness in the 14:00 — 15:30 window, compared with **38%** of their teachers. When a teacher says in the staff room, *"I don't understand why they can't stay awake — I'm fine"*, he is, biologically, telling the truth. The two clocks are just set differently.

⑥ **Sleep debt makes the dip worse.** If a student got only 6.5 hours of sleep instead of the recommended 8.5, the afternoon dip can deepen by **40 — 70%**, not 20%. Combine a late night of phone use, a very carbohydrate-heavy school lunch, and a slightly cold classroom, and the dip becomes a cliff. This is not weakness. It is the sum of three stacked biological signals. The antidote is simple: regular sleep.

⑦ **What actually helps — the honest list.** Dozens of studies across three decades have repeatedly shown the same five useful tricks. **First**, the famous **"coffee nap"** (a teacher's-lounge trick, not for students): drink one small coffee, then rest with eyes closed for 15—20 minutes. Caffeine takes about 20 minutes to reach the brain, so when you wake, you are sharper than by coffee alone. **Second**, a **10-minute walk in daylight** after lunch. Bright light on the skin and eyes quickly raises alertness. **Third**, a short **"power nap" of 10—20 minutes** (not longer, otherwise you enter deep sleep and feel worse). **Fourth**, a **light** lunch — less refined carbs, more protein, some green vegetables. **Fifth**, simply **accept the dip and schedule light, active tasks** in the 14:00 — 15:30 window: group talk, physical activities, note-organising, rather than difficult new theory. A 2024 Finnish study of 22 middle schools rearranged timetables to place harder subjects in the morning and more interactive subjects in the early afternoon; end-of-year test scores rose on average **by 6%**, with no other change.

⑧ **What does not help, despite popular belief.** Fighting the dip with heavy energy drinks between 15:00 and 17:00 produces, on average, a **poorer night's sleep** and **deeper dip the next day**. The body takes revenge. Splashing cold water on the face produces about **90 seconds** of extra alertness and then fades. Standing up abruptly, shaking your head vigorously, or pinching yourself — these look active but give less than three minutes. The afternoon dip does not respond to drama. It responds to respect and small adjustments.

⑨ If there is one sentence to carry home from this article, it is this: *the afternoon sleepiness that quietly catches you between 14:00 and 15:30 is not a character flaw and is not caused only by rice. It is a 50-million-year-old human feature.* Understanding it turns it from a shame into a tool. **Do your hardest thinking in the morning. Be honest with the early afternoon. Sleep enough the night before. Let the dip pass on a short**

walk or a short nap. Then the late afternoon — which, in fact, contains your second peak of alertness, around 17:00 — is quietly yours.

注释： circadian 昼夜节律的；suprachiasmatic nucleus 视交叉上核；tryptophan 色氨酸；serotonin 血清素；sleep debt 睡眠债

根据短文内容，选择最佳答案。

1. According to Paragraph ②, what is the main cause of the afternoon sleepiness?

A. Only a heavy lunch.
B. The circadian rhythm, controlled by the suprachiasmatic nucleus in the brain, which has two fixed dips in its alertness curve per day — a deep one at around 3:00 – 4:00 in the morning and a second, smaller one at around 14:00 – 16:00 in the afternoon; both dips exist in people who skip lunch and even in people isolated for months in Arctic caves.
C. Weakness of mind.
D. Loud classrooms.

2. What did the 2022 meta-study from the University of Loughborough find about lunch's role?

A. Lunch has no effect at all.
B. Lunch completely controls the dip.
C. A lunch heavy in refined carbohydrates deepened the afternoon dip by about 20%, while a light lunch reduced but did not remove the dip — lunch "turns the volume up, not on."
D. Skipping lunch makes the dip disappear.

3. According to the 2023 Beijing Normal University study of 1,740 Chinese middle-school students, what did it find?

A. Teachers were sleepier than students.
B. 73% of Grade 7–9 students reported significant sleepiness between 14:00 and 15:30, compared with 38% of their teachers — because the teenage circadian rhythm runs about 1–2 hours later, the afternoon dip is deeper and earlier in teenagers than in adults.
C. Only girls felt sleepy.
D. Students were only tired on Monday.

4. Which of the following is NOT on the writer's "honest list" of what actually helps with the dip?

A. A "coffee nap" for teachers (drinking coffee, then resting 15–20 min with eyes closed).
B. A 10-minute walk in daylight after lunch.

C. Splashing cold water on the face repeatedly as a long-term solution.
D. A short "power nap" of 10—20 minutes.

5. What is the main point of the final paragraph?

A. The afternoon dip is a sign of laziness.
B. The afternoon sleepiness between 14:00 and 15:30 is not a character flaw and is not caused only by rice — it is a 50-million-year-old human feature; understanding it turns it from a shame into a tool, so one should do hard thinking in the morning, be honest with the early afternoon (walk or short nap), sleep enough the night before, and then the second alertness peak around 17:00 is quietly yours.
C. Students should skip afternoon classes.
D. Coffee solves everything.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段昼夜节律 + SCN + 双谷
【解析】"SCN / 两谷 / 北极洞穴"直引。A、C、D 均错。故选 B。

2.
【答案】C
【题型】细节理解题
【定位】第③段 Loughborough 研究
【解析】"20% / 轻食减少不消除 / 音量"直引。A、B、D 均错。故选 C。

3.
【答案】B
【题型】细节理解题
【定位】第⑤段北师大研究
【解析】"73% / 38% / 1—2 小时晚"直引。A、C、D 均错。故选 B。

4.
【答案】C
【题型】细节理解题
【定位】第⑦⑧段对比
【解析】A、B、D 都在有用列表；C 在"无用"段中。故选 C。

5.
【答案】B
【题型】主旨大意题

【定位】第⑨段结尾
【解析】"5000 万年 + 尊重 + 第二峰"主旨。A、C、D 均偏离。故选 B。